My Design: Adaptive Technology
Armin Zarbaft, Jared Wagner, John Estock

## **Element A: Presentation and Justification of Problem**

Problem Statement:

A special needs classroom in an elementary school has the function of nurturing and teaching special needs students. At all times, the students have an aid or teacher supervising them and helping them with their individual needs. Sensory devices play an important role in the classroom environment by fostering growth, learning, and calmness. In the classroom is a sensory space where various devices and files are stored. The teachers have amassed a myriad of sensory devices, some of which are not in use for the reason that they are often specialized to the needs and wants of individuals. This creates a need for any future sensory devices to be interchangeable and have a proper storage place in the event they are not in use. Adaptability is an important component of the classroom, thus making it important that there are no major changes to infrastructure. Still, there could be benefits in the optimization of the space available, including organizing existing files. One space in particular that can benefit from optimization is the island in front of the sensory room.
The fire marshal has certain restrictions on what can be done to the shelves and bolted furniture in the space. One is that the island at the entrance of the space cannot be taken down or made taller than its current height of 48in.

In a broader context, solving this problem can positively impact other stakeholders, such as school administrators, educators, and even the students' families. An optimized sensory room can potentially serve as a model for other special needs classrooms, contributing to improved learning experiences and outcomes for special needs students in various educational settings. The consequences of not addressing this issue could lead to continued inefficiency in resource utilization and hindered educational support for special needs students in similar environments.

Primary Stakeholders:

Children
Adults in the room (teachers, nurse)

Secondary Stakeholders:

Parents
School staff
Fire marshall
Special needs kids in other schools

Concerns:

- Due to the need for special needs children to have something that prevents and combats overstimulation, the sensory device should appeal to their senses in a calming and satisfying way.
- Due to the need for teachers to be able to easily plan and organize lessons for the kids, the space should be controlled and organized.
- Due to the need for the teachers to keep the kids from messing with documents, files, and other storage not intended for a child's use, optimization of space should include covered storage.
- Due to the need for the special needs children to be engaged in what they are doing, the sensory devices should be prioritized in the immediate display of the sensory room.
- Due to the need of the children not being overwhelmed while in the sensory room, the space should be cleared of clutter
- Due to the need of the adults in the room to enter and move around the space to help the children and access any storage, the space should be optimized for walking room.

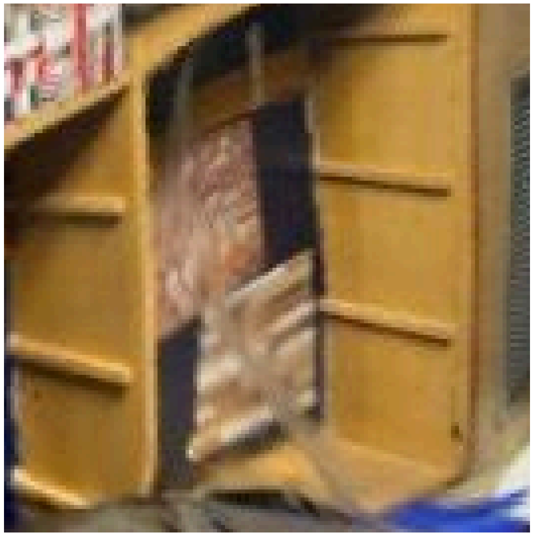
Problem Justification:

- Sensory products have been proven to help people with special needs by engaging their senses and providing calming effects
  - “Sensory Products Archives.” *Enabling Devices*, enablingdevices.com/product-category/sensory/. Accessed 13 Dec. 2023.
- According to *Autism Speaks*, people with autism can have both hypersensitivity and hyposensitivity, with different levels of sensory responsiveness. Visual supports, tactile fidgets, spacious rooms, and weighted apparel have all been proven to help with accommodations for hypersensitivity.
  - “Sensory Issues.” *Autism Speaks*, www.autismspeaks.org/sensory-issues. Accessed 14 Dec. 2023.
- According to the *National Library of Medicine,* when autistic children can control the sensory equipment, they pay more attention and perform fewer repetitive and sensory behaviors.
  - Unwin, Katy L, et al. “The Use of Multi-Sensory Environments with Autistic Children: Exploring the Effect of Having Control of Sensory Changes.” *Autism : The International Journal of Research and Practice*, U.S. National Library of Medicine, Aug. 2022, www.ncbi.nlm.nih.gov/pmc/articles/PMC9340127/.
- Autistic children tend to have sensory overload that is worsened from disorganized spaces. Sensory overload can distract and derail focus. According to *Healing Haven,* organized spaces can help promote productivity and smoother transitions between activities, allowing autistic children to acquire new skills and have the ability to predict what they will be doing. Organization also helps children become more independent and less reliant on additional support.
  - CMotzkus, and Shammy Peterson. “The Benefits of Organized Space for Individuals with Autism.” *Healing Haven*, 30 Aug. 2023, thehealinghaven.net/benefits-of-organized-space-for-individuals-with-autism/#:~:text=Designating%20organized%20areas%20helps%20individuals,smoother%20transitions%20to%20new%20activities.
- Reorganizing spaces by removing clutter can make it easier for children with autism to find and use items
  - “Improving Organizational Skills of Children with Autism.” *Healis Autism Centre*, Healis Autism Centre, 29 Aug. 2023, www.healisautism.com/post/improving-organizational-skills-children-autism.

- Below is an image of the current sensory space:

As seen in the photograph above, there is a lot of clutter and disarray, making it difficult to move around the room and use the various devices. This is some indication that the people who use that room could benefit from additional organization and adjustments to current sensory activities.

Below is a zoomed in image of the current sensory wall:



As seen in the picture above, the sensory wall consists of two similarly sized panels on the back wall of a cubby. This is a fairly simple design, but suffers from its lack of interesting qualities. There could be a benefit in creating a more attractive and adaptable sensory wall.

<u>Summary:</u>

Creating a sensory product that is capable of fostering positive sensory experiences can aid with both hypersensitivity and hyposensitivity, providing serene feelings for its user. Making this product interactive and interesting shifts the user's focus onto it, maximizing the benefits it can have. In order to encourage use of new and existing sensory tools, the sensory space containing these tools should be reorganized in an aesthetically pleasing and efficient manner aimed at optimizing the space.

The first step in developing a sensory device is selecting one or multiple senses the device will be aimed at featuring. Seeing as that the sensory room currently has an insufficient amount of tactile devices, represented by three uniform panels, and that other groups will be working on devices that handle the other senses, we will be setting our focus on improving the tactile devices in the sensory room through creation and improvements.

Through our research, it has been made clear that organization can not only be helpful to the staff working in the classroom as means of productivity and efficiency, but also helpful in hindering sensory overload in the children that is harmful to their learning experience.

In essence, we will be organizing the existing sensory space which includes the addition of tactiles.